

```
[cloudera@quickstart ~]$ hive
```

Logging initialized using configuration in file:/etc/hive/conf.dist/hive-log4j.properties

WARNING: Hive CLI is deprecated and migration to Beeline is recommended.

```
hive> create database mydb2;
```

OK

Time taken: 2.732 seconds

```
hive> use mydb2;
```

OK

Time taken: 0.277 seconds

```
hive> create table flight(fno int, year int, dest varchar(10), delay float);
```

OK

Time taken: 1.514 seconds

```
hive> alter table flight rename to air_flight;
```

OK

Time taken: 0.719 seconds

```
hive> alter table air_flight add columns(source varchar(10));
```

OK

Time taken: 0.298 seconds

```
hive> drop table flight;
```

OK

Time taken: 0.072 seconds

```
hive> desc air_flight;
```

OK

fno	int
-----	-----

year	int
------	-----

dest	varchar(10)
------	-------------

delay	float
-------	-------

source	varchar(10)
--------	-------------

Time taken: 0.633 seconds, Fetched: 5 row(s)

```
hive> create table flight(fno int, year int, dest varchar(10), delay float)
```

```
> row format delimited
```

```
> fields terminated by ','
```

```
> lines terminated by '\n'
```

```
> stored as textfile;
```

OK

Time taken: 0.168 seconds

```
hive> insert into flight values(123,2009,"mumbai",30.6);
```

Query ID = cloudera_20260319062828_34ad2aec-141a-4825-9715-3a4bff7d5eb6

Total jobs = 3

Launching Job 1 out of 3

Number of reduce tasks is set to 0 since there's no reduce operator

Starting Job = job_1773910366565_0002, Tracking URL =

http://quickstart.cloudera:8088/proxy/application_1773910366565_0002/

Kill Command = /usr/lib/hadoop/bin/hadoop job -kill job_1773910366565_0002
Hadoop job information for Stage-1: number of mappers: 1; number of reducers: 0
2026-03-19 06:28:56,846 Stage-1 map = 0%, reduce = 0%
2026-03-19 06:29:24,093 Stage-1 map = 100%, reduce = 0%, Cumulative CPU 4.02 sec
MapReduce Total cumulative CPU time: 4 seconds 20 msec
Ended Job = job_1773910366565_0002
Stage-4 is selected by condition resolver.
Stage-3 is filtered out by condition resolver.
Stage-5 is filtered out by condition resolver.
Moving data to: hdfs://quickstart.cloudera:8020/user/hive/warehouse/mydb2.db/flight/.hive-staging_hive_2026-03-19_06-28-16_415_3959057476386415884-1/-ext-10000
Loading data to table mydb2.flight
Table mydb2.flight stats: [numFiles=1, numRows=1, totalSize=21, rawDataSize=20]
MapReduce Jobs Launched:
Stage-Stage-1: Map: 1 Cumulative CPU: 4.02 sec HDFS Read: 4157 HDFS Write: 89
SUCCESS
Total MapReduce CPU Time Spent: 4 seconds 20 msec
OK
Time taken: 71.462 seconds

hive> insert into flight values(124,2008,"pune",50.6);
Query ID = cloudera_20260319063030_b4a0019f-4c16-4226-ab29-30ae9c2f8fb6
Total jobs = 3
Launching Job 1 out of 3
Number of reduce tasks is set to 0 since there's no reduce operator
Starting Job = job_1773910366565_0003, Tracking URL =
http://quickstart.cloudera:8088/proxy/application_1773910366565_0003/
Kill Command = /usr/lib/hadoop/bin/hadoop job -kill job_1773910366565_0003
Hadoop job information for Stage-1: number of mappers: 1; number of reducers: 0
2026-03-19 06:30:50,203 Stage-1 map = 0%, reduce = 0%
2026-03-19 06:31:47,920 Stage-1 map = 100%, reduce = 0%, Cumulative CPU 6.56 sec
MapReduce Total cumulative CPU time: 6 seconds 560 msec
Ended Job = job_1773910366565_0003
Stage-4 is selected by condition resolver.
Stage-3 is filtered out by condition resolver.
Stage-5 is filtered out by condition resolver.
Moving data to: hdfs://quickstart.cloudera:8020/user/hive/warehouse/mydb2.db/flight/.hive-staging_hive_2026-03-19_06-30-23_032_4841385862567102068-1/-ext-10000
Loading data to table mydb2.flight
Table mydb2.flight stats: [numFiles=2, numRows=2, totalSize=40, rawDataSize=38]
MapReduce Jobs Launched:
Stage-Stage-1: Map: 1 Cumulative CPU: 6.56 sec HDFS Read: 4251 HDFS Write: 87
SUCCESS
Total MapReduce CPU Time Spent: 6 seconds 560 msec
OK
Time taken: 88.117 seconds

```
hive> select * from flight;
OK
123    2009    mumbai    30.6
124    2008    pune     50.6
Time taken: 0.325 seconds, Fetched: 2 row(s)
```

```
hive> load data local inpath "f.txt"
> overwrite into table flight;
Loading data to table mydb2.flight
Table mydb2.flight stats: [numFiles=1, numRows=0, totalSize=113, rawDataSize=0]
OK
Time taken: 0.833 seconds
```

```
hive> select * from flight;
OK
923    2009    navi mumba  60.0
156    2009    kolhapur    78.0
112    2009    pune       0.0
322    2008    nagpur     90.0
132    2008    aurangabad  67.0
Time taken: 0.125 seconds, Fetched: 5 row(s)
```

```
hive> create table nflight(fno int, year int, source varchar(10))
> row format delimited
> fields terminated by ','
> lines terminated by '\n'
> stored as textfile;
OK
Time taken: 0.201 seconds
```

```
hive> insert into nflight values(923,2009,"pune");
Query ID = cloudera_20260319070808_6a837d11-854d-4a5a-b8c0-bcc12e1316eb
Total jobs = 3
Launching Job 1 out of 3
Number of reduce tasks is set to 0 since there's no reduce operator
Starting Job = job_1773910366565_0004, Tracking URL =
http://quickstart.cloudera:8088/proxy/application_1773910366565_0004/
Kill Command = /usr/lib/hadoop/bin/hadoop job -kill job_1773910366565_0004
Hadoop job information for Stage-1: number of mappers: 1; number of reducers: 0
2026-03-19 07:09:05,558 Stage-1 map = 0%, reduce = 0%
2026-03-19 07:09:29,418 Stage-1 map = 100%, reduce = 0%, Cumulative CPU 3.7 sec
MapReduce Total cumulative CPU time: 3 seconds 700 msec
Ended Job = job_1773910366565_0004
Stage-4 is selected by condition resolver.
Stage-3 is filtered out by condition resolver.
Stage-5 is filtered out by condition resolver.
```

Moving data to: hdfs://quickstart.cloudera:8020/user/hive/warehouse/mydb2.db/nflight/.hive-staging_hive_2026-03-19_07-08-38_078_4375339367269004138-1/-ext-10000

Loading data to table mydb2.nflight

Table mydb2.nflight stats: [numFiles=1, numRows=1, totalSize=14, rawDataSize=13]

MapReduce Jobs Launched:

Stage-Stage-1: Map: 1 Cumulative CPU: 3.7 sec HDFS Read: 3894 HDFS Write: 83
SUCCESS

Total MapReduce CPU Time Spent: 3 seconds 700 msec

OK

Time taken: 54.368 seconds

hive> select a.fno, a.year, a.dest,a.delay,b.source

> from flight a join nflight b

> on(a.fno=b.fno);

Query ID = cloudera_20260319071212_ed71f372-d003-4337-8d15-e55255f6d95e

Total jobs = 1

Execution log at: /tmp/cloudera/cloudera_20260319071212_ed71f372-d003-4337-8d15-e55255f6d95e.log

2026-03-19 07:12:14 Starting to launch local task to process map join; maximum memory = 1013645312

2026-03-19 07:12:21 Dump the side-table for tag: 1 with group count: 1 into file:
file:/tmp/cloudera/84dec281-62ab-428f-b0ac-be8703c0bdffa/hive_2026-03-19_07-12-01_809_2415796022714336285-1/-local-10003/HashTable-Stage-3/MapJoin-mapfile01--.hashtable

2026-03-19 07:12:21 Uploaded 1 File to: file:/tmp/cloudera/84dec281-62ab-428f-b0ac-be8703c0bdffa/hive_2026-03-19_07-12-01_809_2415796022714336285-1/-local-10003/HashTable-Stage-3/MapJoin-mapfile01--.hashtable (286 bytes)

2026-03-19 07:12:21 End of local task; Time Taken: 7.082 sec.

Execution completed successfully

MapredLocal task succeeded

Launching Job 1 out of 1

Number of reduce tasks is set to 0 since there's no reduce operator

Starting Job = job_1773910366565_0005, Tracking URL =

http://quickstart.cloudera:8088/proxy/application_1773910366565_0005/

Kill Command = /usr/lib/hadoop/bin/hadoop job -kill job_1773910366565_0005

Hadoop job information for Stage-3: number of mappers: 1; number of reducers: 0

2026-03-19 07:12:48,530 Stage-3 map = 0%, reduce = 0%

2026-03-19 07:13:16,174 Stage-3 map = 100%, reduce = 0%, Cumulative CPU 4.02 sec

MapReduce Total cumulative CPU time: 4 seconds 20 msec

Ended Job = job_1773910366565_0005

MapReduce Jobs Launched:

Stage-Stage-3: Map: 1 Cumulative CPU: 4.02 sec HDFS Read: 6764 HDFS Write: 30
SUCCESS

Total MapReduce CPU Time Spent: 4 seconds 20 msec

OK

923 2009 navi mumba 60.0 pune
Time taken: 76.681 seconds, Fetched: 1 row(s)

```
hive> create index flight_index on table flight(fno)
> as 'org.apache.hadoop.hive.ql.index.compact.CompactIndexHandler'
> WITH DEFERRED REBUILD;
OK
Time taken: 1.095 seconds
hive> show tables;
OK
air_flight
flight
mydb2__flight_flight_index__
nflight
values__tmp__table__1
values__tmp__table__2
values__tmp__table__3
Time taken: 0.141 seconds, Fetched: 7 row(s)
hive>
```

On Flights.csv:-

```
[cloudera@quickstart ~]$ hadoop fs -mkdir /user/hive/flight_data
[cloudera@quickstart ~]$ hadoop fs -put /home/cloudera/Desktop/flights.txt
/user/hive/flight_data
[cloudera@quickstart ~]$ hadoop fs -ls /user/hive/flight_data
Found 1 items
-rw-r--r-- 1 cloudera hive 2495 2026-03-19 08:20 /user/hive/flight_data/flights.txt
[cloudera@quickstart ~]$ hive
```

Logging initialized using configuration in file:/etc/hive/conf.dist/hive-log4j.properties
WARNING: Hive CLI is deprecated and migration to Beeline is recommended.

```
hive> use flights_db;
OK
Time taken: 1.151 seconds
hive> create table flights (year int, month int, day int, dep_delay int ,origin string, dest string>
> ROW FORMAT DELIMITED
> FIELDS TERMINATED BY ',';
OK
Time taken: 1.01 seconds

hive> load data inpath '/user/hive/flight_data/flights.txt' into table flights;
Loading data to table flights_db.flights
Table flights_db.flights stats: [numFiles=1, totalSize=2495]
OK
Time taken: 2.035 seconds
```

```
hive> select * from flights limit 10;
OK
NULL NULL NULL NULL NULL NULL
1949 NULL 112 NULL NULL NULL
1949 NULL 118 NULL NULL NULL
1949 NULL 132 NULL NULL NULL
1949 NULL 129 NULL NULL NULL
1949 NULL 121 NULL NULL NULL
1949 NULL 135 NULL NULL NULL
1949 NULL 148 NULL NULL NULL
1949 NULL 148 NULL NULL NULL
1949 NULL 136 NULL NULL NULL
Time taken: 1.471 seconds, Fetched: 10 row(s)
```

```
hive> insert into flights values (2008,1,3,20,'DEL','MUM');
Query ID = cloudera_20260319082929_cb393387-aed6-487c-a45a-99d8575a7676
Total jobs = 3
Launching Job 1 out of 3
Number of reduce tasks is set to 0 since there's no reduce operator
Starting Job = job_1773910366565_0006, Tracking URL =
http://quickstart.cloudera:8088/proxy/application_1773910366565_0006/
Kill Command = /usr/lib/hadoop/bin/hadoop job -kill job_1773910366565_0006
Hadoop job information for Stage-1: number of mappers: 1; number of reducers: 0
2026-03-19 08:30:14,877 Stage-1 map = 0%, reduce = 0%
2026-03-19 08:30:47,747 Stage-1 map = 100%, reduce = 0%, Cumulative CPU 4.07 sec
MapReduce Total cumulative CPU time: 4 seconds 70 msec
Ended Job = job_1773910366565_0006
Stage-4 is selected by condition resolver.
Stage-3 is filtered out by condition resolver.
Stage-5 is filtered out by condition resolver.
Moving data to:
hdfs://quickstart.cloudera:8020/user/hive/warehouse/flights_db.db/flights/.hive-
staging_hive_2026-03-19_08-29-35_968_2389512136508296799-1/-ext-10000
Loading data to table flights_db.flights
Table flights_db.flights stats: [numFiles=2, numRows=1, totalSize=2515, rawDataSize=19]
MapReduce Jobs Launched:
Stage-Stage-1: Map: 1 Cumulative CPU: 4.07 sec HDFS Read: 4429 HDFS Write: 94
SUCCESS
Total MapReduce CPU Time Spent: 4 seconds 70 msec
OK
Time taken: 75.221 seconds
```

```
hive> alter table flights add columns (airline string);
OK
Time taken: 0.38 seconds
```

```
hive> create EXTERNAL TABLE flights_ext (year int, month int, day int, dep_delay int, origin string, dest string)
```

```
> ROW FORMAT DELIMITED
```

```
> FIELDS TERMINATED BY ','
```

```
> LOCATION '/user/hive/flight_data';
```

```
OK
```

```
Time taken: 0.169 seconds
```

```
hive> create table airport (code string,city string);
```

```
OK
```

```
Time taken: 0.125 seconds
```

```
hive> insert into airport values ('JFK','NewYork'), ('LAX','LosAngeles');
```

```
Query ID = cloudera_20260319083434_d1cad105-9f40-4bc1-b982-6230ca180859
```

```
Total jobs = 3
```

```
Launching Job 1 out of 3
```

```
Number of reduce tasks is set to 0 since there's no reduce operator
```

```
Starting Job = job_1773910366565_0007, Tracking URL =
```

```
http://quickstart.cloudera:8088/proxy/application_1773910366565_0007/
```

```
Kill Command = /usr/lib/hadoop/bin/hadoop job -kill job_1773910366565_0007
```

```
Hadoop job information for Stage-1: number of mappers: 1; number of reducers: 0
```

```
2026-03-19 08:34:46,332 Stage-1 map = 0%, reduce = 0%
```

```
2026-03-19 08:35:12,669 Stage-1 map = 100%, reduce = 0%, Cumulative CPU 2.52 sec
```

```
MapReduce Total cumulative CPU time: 2 seconds 520 msec
```

```
Ended Job = job_1773910366565_0007
```

```
Stage-4 is selected by condition resolver.
```

```
Stage-3 is filtered out by condition resolver.
```

```
Stage-5 is filtered out by condition resolver.
```

```
Moving data to:
```

```
hdfs://quickstart.cloudera:8020/user/hive/warehouse/flights_db.db/airport/.hive-
```

```
staging_hive_2026-03-19_08-34-15_791_3230838328987584713-1/-ext-10000
```

```
Loading data to table flights_db.airport
```

```
Table flights_db.airport stats: [numFiles=1, numRows=2, totalSize=27, rawDataSize=25]
```

```
MapReduce Jobs Launched:
```

```
Stage-Stage-1: Map: 1 Cumulative CPU: 2.52 sec HDFS Read: 3365 HDFS Write: 101  
SUCCESS
```

```
Total MapReduce CPU Time Spent: 2 seconds 520 msec
```

```
OK
```

```
Time taken: 60.008 seconds
```

```
hive> select f.origin, a.city from flights f join airport a on f.origin = a.code;
```

```
Query ID = cloudera_20260319083535_37e9ba6e-5649-4363-b11d-eb84f978b438
```

```
Total jobs = 1
```

```
Execution log at: /tmp/cloudera/cloudera_20260319083535_37e9ba6e-5649-4363-b11d-  
eb84f978b438.log
```

2026-03-19 08:36:07 Starting to launch local task to process map join; maximum memory = 1013645312

2026-03-19 08:36:16 Dump the side-table for tag: 1 with group count: 2 into file: file:/tmp/cloudera/abed5c43-220c-45b7-9ad3-a6156f895798/hive_2026-03-19_08-35-55_144_9090838058344911935-1/-local-10003/HashTable-Stage-3/MapJoin-mapfile01--.hashtable

2026-03-19 08:36:16 Uploaded 1 File to: file:/tmp/cloudera/abed5c43-220c-45b7-9ad3-a6156f895798/hive_2026-03-19_08-35-55_144_9090838058344911935-1/-local-10003/HashTable-Stage-3/MapJoin-mapfile01--.hashtable (323 bytes)

2026-03-19 08:36:16 End of local task; Time Taken: 8.476 sec.

Execution completed successfully

MapredLocal task succeeded

Launching Job 1 out of 1

Number of reduce tasks is set to 0 since there's no reduce operator

Starting Job = job_1773910366565_0008, Tracking URL =

http://quickstart.cloudera:8088/proxy/application_1773910366565_0008/

Kill Command = /usr/lib/hadoop/bin/hadoop job -kill job_1773910366565_0008

Hadoop job information for Stage-3: number of mappers: 1; number of reducers: 0

2026-03-19 08:36:55,680 Stage-3 map = 0%, reduce = 0%

2026-03-19 08:37:24,225 Stage-3 map = 100%, reduce = 0%, Cumulative CPU 3.87 sec

MapReduce Total cumulative CPU time: 3 seconds 870 msec

Ended Job = job_1773910366565_0008

MapReduce Jobs Launched:

Stage-Stage-3: Map: 1 Cumulative CPU: 3.87 sec HDFS Read: 8874 HDFS Write: 0

SUCCESS

Total MapReduce CPU Time Spent: 3 seconds 870 msec

OK

Time taken: 91.388 seconds

hive> create index idx_delay on table flights (dep_delay) as 'COMPACT' WITH DEFERRED REBUILD;

OK

Time taken: 0.579 seconds

hive> alter index idx_delay on flights REBUILD;

Query ID = cloudera_20260319084141_ef22621a-f14e-4cf9-bd5e-05a728de16d3

Total jobs = 1

Launching Job 1 out of 1

Number of reduce tasks not specified. Estimated from input data size: 1

In order to change the average load for a reducer (in bytes):

set hive.exec.reducers.bytes.per.reducer=<number>

In order to limit the maximum number of reducers:

set hive.exec.reducers.max=<number>

In order to set a constant number of reducers:

set mapreduce.job.reduces=<number>

Starting Job = job_1773910366565_0009, Tracking URL =
http://quickstart.cloudera:8088/proxy/application_1773910366565_0009/
Kill Command = /usr/lib/hadoop/bin/hadoop job -kill job_1773910366565_0009
Hadoop job information for Stage-1: number of mappers: 1; number of reducers: 1
2026-03-19 08:41:28,742 Stage-1 map = 0%, reduce = 0%
2026-03-19 08:41:52,879 Stage-1 map = 100%, reduce = 0%, Cumulative CPU 2.62 sec
2026-03-19 08:42:16,652 Stage-1 map = 100%, reduce = 100%, Cumulative CPU 5.64 sec
MapReduce Total cumulative CPU time: 5 seconds 640 msec
Ended Job = job_1773910366565_0009
Loading data to table flights_db.flights_db__flights_idx_delay__
Table flights_db.flights_db__flights_idx_delay__ stats: [numFiles=1, numRows=2,
totalSize=836, rawDataSize=834]
MapReduce Jobs Launched:
Stage-Stage-1: Map: 1 Reduce: 1 Cumulative CPU: 5.75 sec HDFS Read: 11351 HDFS
Write: 935 SUCCESS
Total MapReduce CPU Time Spent: 5 seconds 750 msec
OK
Time taken: 80.09 seconds

hive> show index on flights;
OK
idx_delay flights dep_delay flights_db__flights_idx_delay__
compact
Time taken: 0.218 seconds, Fetched: 1 row(s)

hive> select day, avg(dep_delay) as avg_delay from flights where year = 2008 group by day;
Query ID = cloudera_20260319084444_de20ace5-b248-4b08-8bfb-f8bc6688d01f
Total jobs = 1
Launching Job 1 out of 1
Number of reduce tasks not specified. Estimated from input data size: 1
In order to change the average load for a reducer (in bytes):
 set hive.exec.reducers.bytes.per.reducer=<number>
In order to limit the maximum number of reducers:
 set hive.exec.reducers.max=<number>
In order to set a constant number of reducers:
 set mapreduce.job.reduces=<number>
Starting Job = job_1773910366565_0010, Tracking URL =
http://quickstart.cloudera:8088/proxy/application_1773910366565_0010/
Kill Command = /usr/lib/hadoop/bin/hadoop job -kill job_1773910366565_0010
Hadoop job information for Stage-1: number of mappers: 1; number of reducers: 1
2026-03-19 08:45:07,550 Stage-1 map = 0%, reduce = 0%
2026-03-19 08:45:34,331 Stage-1 map = 100%, reduce = 0%, Cumulative CPU 3.64 sec
2026-03-19 08:45:59,124 Stage-1 map = 100%, reduce = 100%, Cumulative CPU 6.85 sec
MapReduce Total cumulative CPU time: 6 seconds 850 msec
Ended Job = job_1773910366565_0010
MapReduce Jobs Launched:

Stage-Stage-1: Map: 1 Reduce: 1 Cumulative CPU: 6.85 sec HDFS Read: 11166 HDFS
Write: 7 SUCCESS
Total MapReduce CPU Time Spent: 6 seconds 850 msec
OK
3 20.0
Time taken: 81.154 seconds, Fetched: 1 row(s)
hive>